

## STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

08 March 2025

Instructions: Attempt any five out of the following six questions. Specify which five you would like to have graded. Students are allowed calculators, books and notes. Internet use and cell phones are prohibited.

**Problem 1.** A Rutgers statistician reads a journal article on

- Systolic Blood Pressure (SBP)
- Diastolic Blood Pressure (DBP)
- And the within patient Difference (SPB-DBP)

from a study of 100 patients. Assume that the distributions of SBP, DBP and SBP-DBP are i.i.d. multivariate normal and that the sample size  $n = 100$  is large enough to use  $Z$ -statistics instead of  $t$ -statistics.

That paper reported for

- SBP:  $\bar{X} = 145$  and the  $Z$ -test two-sided  $p$ -value for  $\mu_S = 140$  is 0.10.
- DBP:  $\bar{X} = 97$  and  $Z$ -test two-sided  $p$ -value for  $\mu_D = 90$  is 0.05.
- SBP-DBP:  $\bar{X} = 48$  and  $Z$ -test two-sided  $p$ -value for  $\mu_{S-D} = 50$  is 0.05.

He must derive standard deviations and covariances for SBP and DBP from this information.

- (1) Begin with the  $p$ -values obtained for the particular null hypotheses. Then, also using the observed sample means determine what the observed sample standard deviations must have been for both SBP and DBP.
- (2) What is the covariance of SBP and DBP cholesterol in a single observation? (Note that  $\text{SBP} = \text{DBP} + \text{SBP-DBP}$ .)

**Problem 2.** You are provided with the following results (on next page) from an All Possible Regressions of  $Y$ , a continuous response variable, on five predictor variables  $X_1, X_2, X_3, X_4, X_5$ :

- (1) If you were to perform a Forward Stepwise Regression using the MSE criterion to arrive at a Final Model, describe the sequence to **Add/Remove** variables as you arrive at the Final Model? Be sure to state your Final Model.
- (2) If you were to perform a Backward Stepwise Regression using the MSE criterion to arrive at a Model with only the intercept, describe the sequence to **Remove** all 5 variables?

- (3) Briefly describe an experimental situation where Backward Stepwise Regression is computationally slower than Forward Stepwise Regression on predictors.
- (4) Compare Backward Stepwise Regression and Forward Stepwise Regression in terms of the potential risk of including unnecessary predictors in the Final Model. Explain your answer.

| Model Number | Predictors in Model       | MSE   |
|--------------|---------------------------|-------|
| 1            | $X_1$                     | 9.68  |
| 2            | $X_3$                     | 10.85 |
| 3            | $X_2$                     | 18.30 |
| 4            | $X_5$                     | 27.47 |
| 5            | $X_4$                     | 37.10 |
| 6            | $X_2, X_3$                | 9.51  |
| 7            | $X_1, X_3$                | 9.58  |
| 8            | $X_1, X_4$                | 9.79  |
| 9            | $X_1, X_5$                | 9.89  |
| 10           | $X_1, X_2$                | 10.00 |
| 11           | $X_3, X_5$                | 10.22 |
| 12           | $X_3, X_4$                | 10.56 |
| 13           | $X_2, X_5$                | 16.85 |
| 14           | $X_2, X_4$                | 17.68 |
| 15           | $X_4, X_5$                | 27.47 |
| 16           | $X_2, X_3, X_5$           | 8.41  |
| 17           | $X_1, X_2, X_3$           | 9.12  |
| 18           | $X_2, X_3, X_4$           | 9.19  |
| 19           | $X_1, X_3, X_4$           | 9.61  |
| 20           | $X_1, X_3, X_5$           | 9.66  |
| 21           | $X_1, X_4, X_5$           | 10.07 |
| 22           | $X_1, X_2, X_4$           | 10.12 |
| 23           | $X_1, X_2, X_5$           | 10.23 |
| 24           | $X_3, X_4, X_5$           | 10.26 |
| 25           | $X_2, X_4, X_5$           | 16.89 |
| 26           | $X_2, X_3, X_4, X_5$      | 8.47  |
| 27           | $X_1, X_2, X_3, X_5$      | 8.65  |
| 28           | $X_1, X_2, X_3, X_4$      | 9.05  |
| 29           | $X_1, X_3, X_4, X_5$      | 9.80  |
| 30           | $X_1, X_2, X_4, X_5$      | 10.43 |
| 31           | $X_1, X_2, X_3, X_4, X_5$ | 8.74  |

**Problem 3.** Let  $X$  be a continuous random variable with the density

$$f_X(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (1) Compute  $\mathbb{E}[X]$ .
- (2) Compute  $\mathbb{P}(1/3 < X < 2/3)$ .
- (3) Find the probability density function  $f_Y(y)$  of  $Y = \sqrt{X}$ .

Now let  $(X, Y)$  have joint density

$$f_{X,Y}(x, y) = \begin{cases} c \cdot (x + y), & 0 \leq x, y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (4) Find the constant  $c$ .
- (5) Compute  $P(X < 1/2 \text{ and } 1/4 < Y < 3/4)$ .

**Problem 4.** The Poisson distribution with parameter  $\lambda$  and pdf  $= \lambda^y \frac{e^{-\lambda}}{y!}$  has a number of special properties. It has mean = variance  $= \lambda$ . Also, the sum of independent Poisson random variables has a Poisson distribution and the parameter is the sum of all the individual parameters.

Assume we have 2 independent random variables  $Y_1, Y_2$  where  $Y_1$  has a Poisson distribution with parameter  $\theta$  and  $Y_2$  has a Poisson distribution with parameter  $2\theta$ . We want to estimate  $\theta$ .

- (1) Show that  $Y_2 - Y_1$  is an unbiased estimator of  $\theta$  and find its variance.
- (2) Find a sufficient statistic for  $\theta$ .
- (3) Find the maximum likelihood estimator (MLE) for  $\theta$ .
- (4) How does the MLE compare with the estimator in part (1)?

**Problem 5.** Let the random variable  $Y$  be one observation from the discrete distribution having probability mass function

$$f(y|\theta) = \theta(1 - \theta)^{y-1} \text{ for } y = 1, 2, 3, \dots \text{ where } 0 < \theta \leq 1.$$

- (1) Find the form of the MP test of  $H_0 : \theta = 1/2$  vs.  $H_a : \theta = 1/3$ .
- (2) In part (1) what is the actual test procedure when the  $\alpha$ -level =  $1/8$ ?
- (3) What is the power of the test you found in (2)?

**Problem 6.** During an after semester review of courses the grade distributions of two sections of an introductory course were compared. The focus was on the number of A, B and C grades and were recorded in the table below. We assume that student registration in these sections was random. The issue is whether or not that there is any indication that the two courses are not evaluating students in the same way. Complete the table and use an appropriate test of the hypothesis of equivalent distributions in the two sections. What does the p-value suggest?

|               |    | grades |    |    | Row Totals |
|---------------|----|--------|----|----|------------|
|               |    | A      | B  | C  |            |
| Section       | I  | 23     | 40 | -  | 100        |
| Section       | II | -      | -  | -  | 80         |
| Column totals |    | 54     | 72 | 54 | 180        |

## Solutions to Master's Exam Spring 2025

### Solution 1.

(1). For SBP, we may compute

$$\sigma_S = \frac{|\bar{X} - \mu_S|}{z_{1-0.10/2}} = \frac{|145 - 140|}{1.645} = 30.3.$$

For DBP, we may compute

$$\sigma_D = \frac{|\bar{X} - \mu_D|}{z_{1-0.05/2}} = \frac{|97 - 90|}{1.96} = 35.7.$$

(2). Since  $\text{SBP-DBP} = \text{SBP} - \text{DBP}$ , we have

$$\text{Var}(\text{SBP-DBP}) = \text{Var}(\text{SBP}) + \text{Var}(\text{DBP}) - 2\text{Cov}(\text{SBP}, \text{DBP}).$$

From (1), we have:

$$\text{Var}(\text{SBP}) = (30.3)^2 = 918.1,$$

$$\text{Var}(\text{DBP}) = (35.7)^2 = 1274.49.$$

We can calculate the standard deviation of SBP-DBP as:

$$\sigma_{S-D} = \frac{|\bar{X} - \mu_{S-D}|}{z_{1-0.05/2}} = \frac{|48 - 50|}{1.96} = 10.2,$$

from which the variance is:

$$\text{Var}(\text{SBP-DBP}) = (10.2)^2 = 104.04.$$

So we may compute

$$104.04 = 918.1 + 1274.49 - 2\text{Cov}(\text{SBP}, \text{DBP}).$$

equivalently,

$$\text{Cov}(\text{SBP}, \text{DBP}) = 1052.25.$$

### Solution 2.

(1). Step 1: add  $X_1$  followed by  $X_3$  followed by  $X_2$  followed by  $X_5$ . Drop  $X_1$ . Adding  $X_4$  increases MSE, so STOP. The Final model includes  $(X_2, X_3, X_5)$ .

(2). The sequence to remove all predictors to arrive at the model that includes only the intercept:

Remove  $X_1$ , then remove  $X_4$ , then remove  $X_5$ , then remove  $X_2$ , then remove the final predictor remaining which is  $X_3$ .

(3). If the number of candidate predictors is large, say 100, and the number of predictors that belong in the Final Model is small, say 5, then

Backward Stepwise Regression would need to evaluate more models than Forward Stepwise Regression.

(4). Backward Stepwise Regression has a greater risk of including unnecessary predictors in the Final Model than Forward Stepwise Regression. Some reasons include:

Forward selection prevents unimportant variables from entering the model in the first place.

Also, Backward Stepwise Regression:

- starts with all important and unimportant predictors included.
- is more affected by multicollinearity.
- may not aggressively remove predictors that are marginally significant.

### Solution 3.

(1). We may compute

$$\mathbb{E}[X] = \int_0^1 x \cdot 3x^2 dx = 3 \int_0^1 x^3 dx = 3 \left[ \frac{x^4}{4} \right]_0^1 = \frac{3}{4}.$$

(2). We may compute

$$P\left(\frac{1}{3} < X < \frac{2}{3}\right) = \int_{\frac{1}{3}}^{\frac{2}{3}} 3x^2 dx = [x^3]_{\frac{1}{3}}^{\frac{2}{3}} = \left(\frac{8}{27}\right) - \left(\frac{1}{27}\right) = \frac{7}{27}.$$

(3). Let  $F_Y(y) = P(Y \leq y) = P(\sqrt{X} \leq y) = P(X \leq y^2)$ . So

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} \int_0^{y^2} 3x^2 dx = 6y^3 \cdot 2y = 12y^4$$

for  $0 \leq y \leq 1$ , and  $f_Y(y) = 0$  otherwise.

(4). Note that

$$1 = \int_0^1 \int_0^1 c(x+y) dx dy = c \int_0^1 \int_0^1 (x+y) dx dy = c \left( \frac{1}{2} + \frac{1}{2} \right) = c.$$

Therefore,  $c = 1$ .

(5). The desired probability equal to

$$\begin{aligned} \int_0^{\frac{1}{2}} \int_{\frac{1}{4}}^{\frac{3}{4}} (x+y) dy dx &= \int_0^{\frac{1}{2}} \left[ xy + \frac{y^2}{2} \right]_{\frac{1}{4}}^{\frac{3}{4}} dx \\ &= \int_0^{\frac{1}{2}} \left( \left( \frac{3x}{4} + \frac{9}{32} \right) - \left( \frac{x}{4} + \frac{1}{32} \right) \right) dx = \frac{1}{16}. \end{aligned}$$

### Solution 4.

(1).  $\mathbb{E}\{Y_2 - Y_1\} = 2\theta - \theta = \theta$  and  $\text{Var}(Y_2 - Y_1) = 2\theta + \theta = 3\theta$ .

(2). The joint pdf of  $Y_1, Y_2$  is

$$\theta^{y_1} \frac{e^{-\theta}}{y_1!} (2\theta)^{y_2} \frac{e^{-2\theta}}{y_2!} = 2^{y_2} \theta^{y_1+y_2} e^{-3\theta} / (y_1! y_2!).$$

By the factorization theorem,  $U = Y_1 + Y_2$  is a sufficient statistic.

(3). Take the log of the pdf to get

$$y_2 \log(2) + (y_1 + y_2) \log(\theta) - 3\theta - \log((y_1! y_2!).$$

Next take the derivative with respect to  $\theta$  and set it = 0 to get  $(y_1 + y_2)/\theta - 3 = 0$ . Thus the MLE is  $(y_1 + y_2)/3$ .

(4).  $\mathbb{E}\{(y_1+y_2)/3\} = (\theta+2\theta)/3 = \theta$  and  $\text{Var}((y_1+y_2)/3) = (\theta+2\theta)/9 = \theta/3$ .

### Solution 5.

(1). By N.P. Lemma the form of the MP test is to reject when

$$\frac{1}{2} \left(1 - \frac{1}{2}\right)^{y-1} \bigg/ \frac{1}{3} \left(1 - \frac{1}{3}\right)^{y-1} = \left(\frac{3}{4}\right)^y < 2k.$$

This is equivalent to  $(\frac{4}{3})^y > \frac{1}{2k}$ . Thus the simplest form of the test is to Reject  $H_0$  when  $y > k^*$ .

(2). For  $\alpha$ -level =  $1/8$ , we need  $k^*$  such that  $P_0(Y > k^*) = 1/8$ . Since, under the null hypothesis  $P_0(Y = y) = (\frac{1}{2})^y$  and  $P_0(Y > 3) = 1 - P_0(Y \leq 3) = 1 - \frac{7}{8}$  it follows that  $k^* = 3$ .

(3). Under the alternative  $P_a(Y = y) = \frac{1}{3}(\frac{2}{3})^{y-1}$  Therefore, the power is  $1 - P_a(Y \leq 3) = 1 - (\frac{1}{3} + \frac{2}{9} + \frac{4}{27}) = 1 - \frac{19}{27} = \frac{8}{27}$ .

### Solution 6.

Using the marginal totals complete the table to get

|               |    | grades |    |    | Row Totals |
|---------------|----|--------|----|----|------------|
|               |    | A      | B  | C  |            |
| Section       | I  | 23     | 40 | 37 | 100        |
| Section       | II | 31     | 32 | 17 | 80         |
| Column totals |    | 54     | 72 | 54 | 180        |

To get the expected cell amount multiply the row total by the column total and divide by the grand total. The first 2 cells in the first row become  $100 \times 54 / 180 = 30$  and  $100 \times 72 / 180 = 40$ . The remaining expected cell amounts can be obtained using the marginal totals. Thus the expected cell counts are

|               |    | grades |    |    | Row Totals |
|---------------|----|--------|----|----|------------|
|               |    | A      | B  | C  |            |
| Section       | I  | 30     | 40 | 30 | 100        |
| Section       | II | 24     | 32 | 24 | 80         |
| Column totals |    | 54     | 72 | 54 | 180        |

Compute the chi-square statistic  $\sum \sum (E_{ij} - O_{ij})^2 / E_{ij} =$   
 $(30-23)^2/30 + (40-40)^2/40 + (30-37)^2/30 + (24-31)^2/24 + (32-32)^2/32 + (24-17)^2/24$   
 $= 1.63 + 0 + 1.63 + 2.04 + 0 + 2.04 = 7.34$ . Using the chi-square table  
 with degrees of freedom  $= (r-1)(c-1) = 1 \times 2 = 2$  shows the p-value to  
 be about 0.025. Thus equality of grading is questionable.